

BEYOND THE SCORE

Overview

Relational Systems Simulation · Pilot Study · Aotearoa New Zealand, 2025

WHAT IS RELATIONAL SYSTEMS SIMULATION?

Relational Systems Simulation (RSS) is a researcher-directed design method that generates anticipatory knowledge about how AI-mediated decision systems will interact with culturally differentiated, relationally organised households — before those interactions produce documented consequences.

The method combines synthetic ethnography, scenario-based design, and large language models used as bounded generative instruments: not autonomous sources of cultural knowledge, but constrained transformers provisioned with a curated research corpus — community-produced research, national datasets, policy frameworks, and scholarly literature — and directed by the researcher. Every cultural and behavioural attribute in an RSS persona carries a recorded provenance status: anchored in the case's trace sheet to a community-produced source or scholarly literature, or marked as constructed where the corpus provides no warrant. The anchoring constrains generation and makes the construction auditable. A post-pilot audit corrected anchors that claimed more than their sources supported, and the trace sheets record those corrections.

Personas are constructions, not descriptions of actual individuals. Every attribute is either anchored to the cited corpus or marked as constructed; none is a claim about any real person or household.

RSS was developed to investigate emotional and cultural life under conditions of systemic pressure — algorithmic decision making, climate-driven displacement, pandemic response, institutional withdrawal — scenarios that cannot ethically be staged with actual participants.

WHY IT MATTERS

AI-mediated systems are already making consequential determinations about housing, credit, insurance, and welfare for populations whose relational, cultural, and economic lives they do not adequately model. A household whose income arrives informally is not financially unstable; it is financially irregular in the specific sense that algorithmic scoring requires PAYE regularity as a proxy for creditworthiness. A family on ancestral land without individual title is not homeless; it is address-unverifiable in the sense that insurance systems require formally registered addresses. These are predictable outputs of systems designed around assumptions that exclude culturally differentiated, collectively organised, and economically precarious populations, rather than malfunctions.

THE PILOT STUDY

The pilot was conducted in the second half of 2025, on the eve of the phased commencement of the Consumer Data Right framework in Aotearoa New Zealand. Four intergenerational family simulations were generated across three generations and twelve personas, then subjected to plausibility and experiential review by five Māori, Pacific, and migrant Asian youth. The AI-mediated decision scenarios modelled — open banking consent interfaces, predictive lending, insurance risk assessment, and climate adaptation — were imminent institutional realities rather than hypothetical futures.

Case	Family	Community and location	Scenario
1	Letoa-Haumata	Samoan-Māori · South Auckland	Open banking consent · climate-driven managed retreat
2	Riley	Pākehā · Palmerston North	Flood insurance withdrawal · overdraft denial
3	Liu	Chinese-Malaysian · East Auckland	Income irregularity flag on youth banking application
4	Tāwhai	Māori · near Kaikohe	Address unverifiable for insurance discount · whenua-based exclusion

CROSS-CASE FINDINGS

Four relational configurations recurred across the four constructed cases, plus one design blind spot distinct in kind. They are properties of the constructed cases and make no claim about any population.

1 · Relational buffering of institutional risk

Across the cases, institutional risk is rendered as something negotiated through kinship rather than through formal support. The buffering is itself misread as financial irregularity, so buffering and algorithmic misclassification reinforce each other.

2 · Institutional misinterpretation of relational behaviour

In the rendered encounters, systems designed around individual, formally documented, economically regular subjects misread relational behaviour the cases present as culturally normative. Multigenerational composition reads as financial complexity; informal income as instability; communal land tenure as address unverifiability; collective decision-making as absent individual agency.

3 · Emergent distrust of automated systems

In the rendered encounters, distrust toward automated systems forms and transmits across generations, taking a different shape at each generational position. Younger characters experience trust betrayal when systems reveal themselves as surveillant;

middle-generation characters hold protective distrust from accumulated experience; elders hold settled distrust rooted in longer histories of displacement.

4 • Latent resilience

The cases render households maintaining relational architectures that conventional risk models cannot register, holding the households together under institutional pressure. One validator recognised his own family’s practice in Popo’s account of money kept hidden.

Design blind spot • emotional labour

In each case as rendered, a household member, in three of the four the youngest, mediates between the household and institutional systems — translating interfaces, navigating consent forms, carrying the emotional cost of exclusion events. This labour appears in no risk score or verification dataset, yet in the cases it is how the households stay connected to systems built for individuals. The mediating role is anchored in the corpus for two of the cases; what the simulations surfaced is its cost. Four validators recognised the role from their own lives.

VALIDATION

Five validators (Māori, Pacific, and migrant Asian youth, ages 21–25) evaluated three of the four simulations in semi-structured interviews, each assessing two cases; four of the five assessed one case close to their own background and one further from it. Responses fell into two methodologically distinct types: plausibility assessments, evaluating believability against community knowledge; and experiential corroboration, in which a validator recognised a simulated dynamic from their own life, demonstrating that the structural conditions mapped were real, not only that the narrative felt believable. Each of the five validators corroborated at least one dynamic this way.

Simulation	Emotional realism: mean	Cultural resonance: mean	Emotional realism: range	Cultural resonance: range
1 • Letoa-Haumata	4.17	3.67	4–4.5	3–4
2 • Riley (excluded)	—	—	—	—
3 • Liu	4.17	4.00	3.5–5	3–5
4 • Tāwhai	4.38	4.38	4–5	4–5

Simulations 1 and 3 were each rated by three validators, Simulation 4 by four; Simulation 2 was not assessed.

WHAT IS IN THIS SET

Each of the four cases is held in two documents. The reading edition carries the household and the simulation script, for reading; the trace sheet carries the case's design position, corpus anchors, and references, for checking.

- Case 1 • Letoa-Haumata — Samoan-Māori family, South Auckland. Open banking and climate-driven managed retreat.

- Case 2 · Riley — Pākehā family, Palmerston North. Flood insurance withdrawal and overdraft denial.
- Case 3 · Liu — Chinese-Malaysian family, East Auckland. Income irregularity flag on a youth banking application.
- Case 4 · Tāwhai — rural Māori family, near Kaikohe. Address unverifiable for an insurance discount.

Case-keying here is by ethnicity, which is how the pilot was designed. Within-group variation exceeds what culture-keyed archetypes can carry, and the cases should be read as particular households under institutional pressure rather than as portraits of communities.

NOTE ON USE

These cases are made publicly available to support transparency in simulation-based design research and to contribute to ongoing conversation about anticipatory governance methods for AI-mediated systems. Personas are composite research models, not representations of specific individuals. Citations were re-verified against their sources in a post-pilot audit, and the trace sheets record the corrections. A journal article describing the RSS method, its provenance architecture, and the cross-case analysis is in development.